Prepared for referee review

NEURAL BELLMAN OPERATORS

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Neural Bellman Operators construct continuation critics and feasible actors for recursive economic decisions. We separate action-aligned regret from absolute-value precision, derive rationally verified finite calculations of the information in complete Bellman supports, and prove continuous-action regret resolution with a sharp grid bound. Exact geometry reuse preserves all decisions and certificates while reducing repeated algebra. Two prospectively frozen studies retain the original continuous-risk investment model, 56 neural and conventional actors, and strong online and dynamic-programming controls. Direct stopping meets the stronger regret target in all 522 service cases, compared with 296 for the fused control. Independent reconstruction certifies a lower mean original cost against that control, but leaves the comparison with warm adaptation unresolved. In a fresh paired study, geometry reuse reduces complete direct-service cost by 51.3 percent; direct service remains 22.2 percent more expensive than equally optimized warm adaptation. Complete-support computation sharpens 48 contrast bounds without producing an additional qualified policy gain. Independent finite-design refitting and exact arithmetic audits delimit each claim. Matched hinge controls and complete-cost adoption comparisons preserve the remaining neural-specific contribution question.

KEYWORDS: Dynamic programming, Neural approximation, Recursive risk, Policy qualification, Computational economics.

1. INTRODUCTION

A firm choosing investment needs the continuation values of feasible actions, the risk of their future consequences, and a decision about how much computation to buy. These three objects are linked but are not interchangeable. A small critic fitting error can conceal the action error that matters for investment. An uncertain common value level can coexist with an exactly identified optimal action. A more

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accurate certificate can leave the implemented policy unchanged. Neural Bellman Operators (NBO) address this recursive economic decision problem by constructing a continuation critic, a feasible actor, and an account of the original policy's error and complete computational cost.

Our running application has nonlinear investment technology, a state-dependent capacity, continuous innovations and a nested tail-risk preference. Optimality is defined by the original Bellman recursion. The innovation bins in a numerical calculation bound its continuous law; they do not define a replacement economy. The same implemented future controller is charged in every method. The central question is whether learned continuation information can improve the original decision, or reduce the complete work required to make it to a stated economic accuracy, relative to strong conventional methods with the same information.

We develop a decision service that separates action accuracy from absolute value accuracy. Previously computed global Bellman supports imply both a constrained derivative enclosure and an interval containing the original optimizer. These quantities can certify small original-policy regret even when a level-gap certificate is unnecessarily wide. The service buys a proposed action and its first original-risk support directly, uses separate action and value gates to determine whether further refinement is needed, and pays for a full conventional fallback when qualification is unsuccessful. A later policy change requires a strictly negative bound on the actual original contrast. A smaller regret certificate for an unchanged action is recorded as better information, not as a new policy gain. The supplement proves the projected-gradient, rounded-bracket, terminal-width and improvement statements used by this implemented rule.

A second contribution makes the complete Bellman support characterization computable with finite verification. Each paid support constrains the objective on every feasible action, with one slope shared across that domain. The retained finite interpolation theorem describes exactly the original functions compatible with these records. We construct outer linear cuts at possibly different rational points and combine their numerical dual proposals with an exact residual correction. Any accepted combination yields a valid action contrast, even when the optimization routine fails or reaches its resource cap. A separate finite-grid theorem turns these

contrasts into a continuous-action regret bound, with a sharp $Lh^2/8$ resolution term and an exact cellwise improvement. A verified primal gap, when available, identifies proximity to the information bound; a solver status does not. Under an explicitly compulsory further computation cost, the remaining-regret bound also gives a sufficient economic stopping rule.

These calculations build on classical cutting planes, smooth convex interpolation and information-based optimization (Kelley, 1960, Taylor et al., 2017, Florea and Nesterov, 2024). Their contribution is the specific, fully verified passage from the original recursive Bellman payload to a feasible action, its economic regret, and a paid decision to continue computing. No general invention of cutting planes is claimed. The full-support solver is compared with a conventional certificate that uses the same derivative cap, supports, curvature, and already acquired original-optimum lower bound. Neural, hinge, spline, polynomial and online representations have the same certification rights under this information.

The original NBO construction remains central. Its critic is constructed backwards using its own acquired continuation; its actor carries a feasible witness. The operator foundations establish existence and uniqueness of the finite-horizon value, measurable selection, stability, boundary conditions and implementation allowances. One-sided and centered recursive-risk accounts propagate local errors to the original objective. The supplement retains the complete actor/critic computational-graph proofs and the conditional link to stochastic control, nonlinear recursive utility and forward–backward systems. Those links require their stated well-posedness, filtration, regularity and boundary assumptions. Evidence under expectation, nested AVaR or a different recursive driver retains that particular economic objective.

The new prospective computation includes all 56 previously acquired actors, covering ReLU and exactly matched ordinary hinges, quadratic, spline, polynomial and grid representations. It also includes the original-model online controller and uniform-action dynamic programming. All nine off-training states, both risk levels, complete method catalogue, precision ladders, budget caps and verification rules were committed remotely before the first evaluation. The three complete services pay their own work. The support-solver study uses a separately declared subset

of those newly executed records, and charges its additional algebra. A further independent finite-design audit rebuilds the acquired feature matrices and fitting problems with separately written code. It audits the historical training inputs without changing the actors deployed by the experiment.

This division of work is substantive. The nine-state study is a deterministic census with no population confidence interval. Independent reconstruction can confirm, contradict or leave unresolved an original-value or regret claim; mere interval overlap is not confirmation. The finite-design fitting audit does not supply fresh economic labels, whole-domain training validation, or an independent implementation of the shared linear-programming library. The capped support solver does not inherit the limiting accuracy of an ideal exact cutting-plane algorithm. Each restriction is tied to a concrete theorem or execution record.

The strongest conventional wins and the unproved neural adoption regions remain part of the evidence. ReLU and its exactly matched native hinge encode the same acquired function and feasible policy. An advantage they share cannot identify a neural-specific effect. Acquisition, fitting, rejected candidates, certification, inference, output and complete decision clocks enter the comparison. The goal remains a competitive neural method for recursive economic decisions; the new algorithms and information bounds sharpen the test that a learned representation must pass. Complete proofs are in the technical supplement. The development companion retains earlier economic models, all prior source statements and negative findings, distinct experimental identities and the preservation map.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the earlier expected-cost experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. Section 5 evaluates the same investment technology under an explicitly nested AVaR preference and retains the original continuous innovation law. The direct-cost inference in Section DP is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It works backwards with its own acquired continuation, returns a neural critic and a feasible actor, and records an economic error allowance. Its continuation may be fitted or enclosed by an explicitly paid primitive Bellman calculation; an exact optimal continuation is never supplied as a free oracle. The constructive theorem below uses its own fitted f_{t+1} at every date. In the new two-date application, the terminal function g is an original primitive, and the paid last-date operator $T_1 g$ supplies outward target intervals shared with every conventional comparator. The deployed last-date policy retains its separately proved error. Arbitrary trained networks can be evaluated by the first result below. The constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \le \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^{\pi} f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^{\pi} - V_0^{*} \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 4.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \tag{5}$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in* (5) *have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$\begin{aligned} -e_t \le f_t - \mathcal{T}_t f_{t+1} &\le D_t k_t + e_t + 2L_t h_t, \\ \mathcal{T}_t^{\pi} f_{t+1} - f_t &\le e_t + D_t \zeta_t, \end{aligned} \tag{6}$$

and hence

$$J_0^{\pi} - V_0^* \le \sum_{t<T} B_t (D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T (2e_T + 2L_g h_T). \tag{7}$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 4.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

4. REFERENCE-CENTERED RISK KERNELS AND COMPLETE-POLICY QUALIFICATION

A datewise worst-case error account can charge incompatible states at successive dates. Under nonlinear recursive preferences, replacing that account by an expectation under the physical transition law is not valid. We instead retain the conditional risk evaluation and center its dual representation at the actual stored NBO continuation. The resulting state-dependent certificate has an original-optimum interpretation and a finite occupation representation. It extends, rather than replaces, the signed qualification of Section 2.

4.1. *The centered operator*

Retain the original augmented economic state, feasible actions, nonnegative discounts, and bounded complete-policy recursions. At each (t, x, a) suppose the original conditional evaluation admits the representation

$$\rho_{t,x,a}(v) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int v(y) q(dy) - \psi_{t,x,a}(q) \right\}. \tag{8}$$

Every q is a probability measure on the original successor state space. The representation is assumed on a common bounded-function domain containing all continuations below, with the measurability and selection conditions needed for the original Bellman recursions. This is a hypothesis about the original preference primitive, not permission to replace a static risk objective by a nested one. Conditional risk duality and risk-averse dynamic programming are established principles (Ruszczyński, 2010, 2014). For a nonconvex or non-cash-invariant aggregator lacking (8), the earlier order and modulus results remain applicable.

Fix one qualified parameter θ and suppress its subscript. Its entire sequence of stored continuations is $f = (f_t)_{t=0}^{T}$, and its supplied complete feasible policy is π. Define, using the same f_{t+1} in every term,

$$r_t(x,a) = c_t(x,a) + \beta_t \rho_{t,x,a}(f_{t+1}) - f_t(x),$$
$$k_t^f(x,a,q) = \psi_{t,x,a}(q) + \rho_{t,x,a}(f_{t+1}) - \int f_{t+1}\, dq,$$
$$\mathcal{E}_{t,x,a}^f(z) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int z\, dq - k_t^f(x,a,q) \right\}. \tag{9}$$

The penalty is nonnegative and has infimum zero, whether or not the supremum in (8) is attained. Direct substitution gives

$$\mathcal{E}_{t,x,a}^f(z) = \rho_{t,x,a}(f_{t+1} + z) - \rho_{t,x,a}(f_{t+1}). \tag{10}$$

In particular, $\mathcal{E}^f(0) = 0$, and $\mathcal{E}^f$ is monotone, cash invariant, and nonexpansive in the uniform norm. Discarding k^f is an upper relaxation, not this identity. Freezing one risk-maximizing kernel at f is in general not even an upper relaxation.

THEOREM 3—State-dependent original-optimum certificate: *Suppose verified functions satisfy $r_t^-(x,a) \le r_t(x,a)$ for every original feasible action and $r_t(x,\pi_t(x)) \le r_t^{\pi,+}(x)$. Let bounded measurable L_t, U_t satisfy*

$$L_T \le g - f_T \le U_T,$$
$$L_t(x) \le \inf_{a \in A_t(x)} \{ r_t^-(x,a) + \beta_t \mathcal{E}_{t,x,a}^f(L_{t+1}) \},$$
$$U_t(x) \ge r_t^{\pi,+}(x) + \beta_t \mathcal{E}_{t,x,\pi_t(x)}^f(U_{t+1}). \tag{11}$$

Then, at every initial state,

$$f_0 + L_0 \le V_0^* \le V_0^\pi \le f_0 + U_0, \qquad 0 \le V_0^\pi - V_0^* \le U_0 - L_0. \tag{12}$$

The same statements hold at every intervening date and state. They are uniform over a qualified parameter class whenever all premises and the reported bound hold for each member with its own common continuation sequence and complete policy. The certificate permits

endogenous states and actions changing at every date; it does not require a finite state space or independent deployment contexts.

With exact residuals and terminal discrepancies, equality recursions in (11) *give* $L_t = V_t^* - f_t$ *and* $U_t = V_t^\pi - f_t$. *With scalar bounds from* (1), *their corresponding constant lower and upper sequences are feasible certificates here. Thus the exact state-dependent recursion is no looser than the signed scalar account. A strictly smaller bound can result without changing a fit, a feasible action, or an economic observation.*

The last assertion concerns the best exact recursion or a verified improvement upon the constant certificates. It is not a claim that every approximately computed L, U improves the scalar account. In implementation, the minimum of valid policy supersolutions and the maximum of valid optimum subsolutions retain validity, so an inferior new bound need not displace the existing one.

The lower inequality requires every original action. A discrete action search without a verified original-menu lower bound cannot establish it. For a fixed action, a feasible risk kernel supplies a lower bound on $\mathcal{E}^f(L)$; an upper bound requires control of the entire risk envelope. Numerical integration, feasible repairs, residual errors and penalty errors belong in these inequalities. If penalty enclosures are used, lower penalty endpoints enter an upper certificate and upper endpoints enter a lower witness. Approximate dual optimizers are accepted only through these independently checked inequalities.

5. CONTINUOUS-LAW QUALIFICATION IN THE INVESTMENT MODEL

The risk theorem becomes a numerical method only when its inequalities can be constructed from economic primitives. We now supply this construction for the original investment technology with a nested tail-risk preference. States and innovations remain continuous, every feasible action is covered, and the implemented terminal controller rather than a hypothetical exact optimizer determines policy cost. This application joins continuation construction, the original-optimum comparison and a representation comparison on one problem. It does not replace the general NBO construction or the other economic applications.

5.1. *Primitives and the object being approximated*

Let $d \geq 2$ be even, $X = [0,1]^d$, and interpret subscripts cyclically. At dates $t = 0, 1$, the feasible investment is $0 \leq a \leq A(x) = 1/8 + \bar{x}/8$, where $\bar{x} = d^{-1} \sum_j x_j$. Write $b_j = 1/2$ for odd j and $b_j = 1/4$ for even j, and $s_j = (-1)^{j+1}$. The original transition is

$$F_j(x,a,z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1-x_{j+1})}{16} + b_j a + s_j z, \qquad z \sim \mathrm{Unif}[-1/32, 1/32]. \tag{13}$$

Define

$$S(x) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2(1/2 - 2\bar{x})_+^2, \tag{14}$$

$$g(x) = S(x) + \frac{2}{d} \sum_j (x_j - 5/8)^2, \qquad h(a) = a^2 + 4a^4.$$

We use $\beta = 15/16$ and, separately, $\alpha \in \{1/2, 3/4\}$. Average value at risk is the upper-tail cost functional

$$\rho_\alpha(v) = \sup \left\{ \int v q \, dP : 0 \leq q \leq (1-\alpha)^{-1}, \int q \, dP = 1 \right\}.$$

Here P is the original uniform innovation law. The objective is recursively defined by $V_2 = g$ and

$$V_t(x) = S(x) + \inf_{0 \leq a \leq A(x)} \{ h(a) + \beta \rho_\alpha(V_{t+1}(F(x,a,\cdot))) \}.$$

It is not the static tail risk of total realized cost. A complete policy value replaces each infimum by the action prescribed by that policy, with the same recursive risk primitive.

Assumptions for the continuous construction. For this subsection, the primitives are exactly (13)–(14); α is fixed before construction; the fitted continuation f is a finite convex function on X; and every numerical lower or upper value is an outward enclosure. Feasible tail densities and feasible actions are retained explicitly. Terminal selection is the measurable, dyadically acquired controller described below, with

its fallback included in the error bound. These concrete conditions imply compact feasible graphs, bounded costs, measurable minimizers, and the conditional-risk dual used in Theorem 3. They do not assert that every recursive preference or every trained network has this convex structure.

PROPOSITION 4—Primitive curvature and state supports: *For either risk level and every even d, the one-date objective $Q_1(x,a) = S(x) + h(a) + \beta\rho_\alpha(g(F(x,a,\cdot)))$ is jointly convex on $X \times [0,1/4]$, with curvature bounded below by*

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2 \tag{15}$$

in a direction (v,w). Its value V_1 is convex on X and satisfies

$$|V_1(x) - V_1(y)| \le \frac{15}{d}\|x - y\|_1. \tag{16}$$

Let W be the objective evaluated under any fixed feasible risk density, at an anchor (x_0,a_0), and let (p_x,p_a) be its gradient there. For any $\lambda,\nu \ge 0$, set $e = p_a + \lambda - \nu$. Then the following is a global affine minorant of V_1:

$$\ell(y) = W + \lambda(a_0 - A(x_0)) - \nu a_0 - \frac{e^2}{4} + \left(p_x - \frac{\lambda}{8d}\mathbf{1}\right)^\top (y - x_0). \tag{17}$$

Neither optimal multipliers nor an exact optimizing action are required.

The curvature is obtained from the original nonlinear technology, not imposed on an interpolated value grid. The negative second-order term from $x_j(1 - x_{j+1})$ is bounded using $|\partial_j g| \le 10/d$; the stage curvature dominates it. The allowance $e^2/4$ pays for imperfect stationarity. The proof in the Technical Supplement also establishes the global state modulus in (16), including the state-dependent action constraint.

5.2. *An all-action certificate under the original innovation law*

Partition the innovation interval into n equal bins B_i, with midpoints z_i and probabilities $p_i = 1/n$. An innovation bin is a set over which an inequality holds,

not a replacement point mass in the economic model. Choose nonnegative $q_i \leq p_i/(1-\alpha)$ summing to one. These masses define a feasible density constant on each bin. For a fixed initial state and feasible proposal a_0, let W_0 be a lower bound on the objective under this density, obtained from globally valid affine supports of the continuation at the successor midpoints. Let p enclose the derivative of the same supporting objective at a_0. Then

$$\underline{V}_0(x) = \inf_{0 \leq u \leq A(x)} \{W_0 + p(u - a_0) + (u - a_0)^2\} \tag{18}$$

is a lower bound on the original Bellman optimum when the supported continuation is V_1. The scalar quadratic is minimized over the full original interval. Interval uncertainty in p is covered by taking the smaller minimum at its two endpoints, since the minimum is concave in p.

For a convex fitted continuation, midpoint supports give the lower risk witness and the larger endpoint value bounds the continuation throughout each bin. For V_1, Proposition 4 supplies midpoint supports and the allowance $15/(32n)$ covers each half-bin. Upper tail risk is the average of the largest $n(1-\alpha)$ upper bin bounds. This average is the exact risk of a dominating step function, hence an upper bound under the continuous law.

THEOREM 5—Constructed original-optimum qualification: *Suppose the supports and bin bounds above are outwardly verified. Suppose further that the complete terminal controller satisfies $V_1 \leq V_1^{\widehat{\pi}} \leq V_1 + \varepsilon_1$ everywhere on X. For every queried initial state, the lower value (18) and the binwise upper bound $\overline{J}_0^{\widehat{\pi}}$ satisfy*

$$0 \leq J_0^{\widehat{\pi}}(x) - V_0(x) \leq \overline{J}_0^{\widehat{\pi}}(x) - \underline{V}_0(x). \tag{19}$$

For any stored convex continuation f, the same lower and upper constructions are admissible bounds for the centered-risk comparison of Theorem 3. Their validity does not depend on the representation of f, on a successful fit, or on the proposal being an exact minimizer.

To see the connection, write a verified continuation bracket as $\ell \leq V_1 \leq u$ and use errors $L = \ell - f$, $U = u - f$. The centered identity gives

$$\rho_\alpha(f) + \mathcal{E}^f(L) = \rho_\alpha(\ell), \qquad \rho_\alpha(f) + \mathcal{E}^f(U) = \rho_\alpha(u).$$

Joint evaluation avoids subtracting and then adding two separately rounded copies of $\rho_\alpha(f)$. The bound (18) uses one feasible risk density to minorize the required all-action infimum; the policy upper bound controls the full risk envelope. Thus the application constructs sufficient inequalities for the original theorem. It does not claim that a non-neural continuation would receive a different validity theorem, or that computing the original-value witness is free.

The terminal controller observes each state coordinate rounded down to 40 binary places. It accepts its floating proposal only after an outward all-action gap of at most 2^{-21} at that observed state; otherwise it uses rational bisection. The original-state transfer is less than $30\,2^{-40}$. Consequently $\varepsilon_1 = 2^{-20}$ covers the complete controller, including observation and fallback. Policy evaluation encloses this controller at every possible successor, not only at the midpoint states used by the integrator.

Well-posed original decisions. The comparison uses a specified conditional nested recursion on the original observation filtration. A sufficient primitive model has a compact state space, nonempty compact feasible sets varying continuously with state, continuous costs and invariant transitions, compact innovations with their original fixed laws, and finite monotone cash-invariant risk evaluations. These assumptions make the Bellman operator preserve continuous functions. For implemented Borel policies, the risk evaluation must also preserve Borel parameter kernels; the supplement proves this property for expectation and AVaR. Proposition S.10 proves, rather than assumes, the resulting finite-horizon value existence and uniqueness, attained Borel minimizers, comparison against all feasible adapted history policies, and the dated stability bounds. These are classical foundations (Ruszczyński, 2010, 2014), not a claim of a new fixed-point principle. Pointwise transition continuity with an unbounded innovation support is insufficient for arbitrary monetary risk; the exact essential-uniform alternative is stated in the proof.

The original investment primitives satisfy these conditions directly: $F(x,a,z) \in [1/32, 27/32]^d$, $1/8 \le A(x) \le 1/4$, and all costs are bounded. Their joint curvature gives V_1 state strong-convexity modulus $(4 - 5\beta/4)/d$ and root action modulus two. Both dates therefore have unique continuous constrained optimal actors. Proposition S.11 also derives the continuous value gradient with its capacity multiplier,

the original-risk root derivative, and the actual observation/feasibility allowance. Differentiating a rounded actor is not used to establish any of these claims. The supplement records the full smooth critic–actor graph and distinguishes its gradient from a frozen-target update.

The separate Brownian recursive-utility bridge has its own existence and verification assumptions. Proposition S.12 supplies the stopped-domain version, charging the original stopping boundary and terminal condition. Its Lipschitz driver can depend on the backward value, in which case cash invariance is not automatic. The continuous-time stability factor and smooth critic graph are therefore verified within that original recursive criterion.

5.3. *Original marginal regularity and the acquired neural critic*

The present computation specializes the preceding investment primitives to $d = 2$. Write $q_x(a) = S(x) + h(a) + \beta\rho_\alpha\{V_1(F(x,a,Z))\}$. Theorem S.4 proves that q_x has a continuous inward derivative on its full feasible interval, is 2-strongly convex, and has derivative Lipschitz constant

$$L_\alpha = 5 + \beta\left\{\frac{5M_\alpha}{16} + \frac{(4M_\alpha/5)^2\alpha}{(181/64)(1-\alpha)}\right\},$$

$$M_\alpha = \frac{1413}{128} + \beta\left\{\frac{127}{16} + \frac{19625\alpha}{20480(1-\alpha)}\right\}.$$

These constants are derived from the original technology and recursive risk, including binding capacity and an adverse tail touching either innovation boundary. They are not estimates of a fitted critic's curvature.

The retained acquired actor uses an anchored continuation of the form

$$\widehat{R}_\theta(x,v) = \widehat{R}_0(x) + \sum_j \theta_j\{\phi_j(x,v) - \phi_j(x,0)\}, \qquad v = a/A(x).$$

Its known stage cost remains analytic. Paid original-risk value enclosures supply either the value loss or outward finite-difference marginal labels. The complete finite-box fitting procedure and exact optimization-gap audits are in Supplement, Section 3. Their off-design feature-cover theorem states the additional information

needed to infer global action accuracy from such training; the earlier experiment did not execute that global cover. The new service below instead acquires and reuses original information at the action actually proposed by the actor.

The resulting complete-policy cost remains

$$J_x(a) = q_x(a) + \eta_x(a), \qquad 0 \le \eta_x(a) \le \delta = \beta 2^{-20}.$$

The allowance is uniform and concerns the original feasible terminal controller. No derivative or convexity of η_x is assumed. A state-dependent value anchor must still be carried when a critic is exported to an earlier Bellman operator: Theorem S.18 gives the full level-and-contrast recursion and its distinct noncash-invariant version. Local action invariance never licenses deleting that recursive level.

6. TURNING A PAID BELLMAN CERTIFICATE INTO A SAFE DECISION

A learned actor can be useful even when its fitted marginal is not a reliable local direction. The original qualification calculation already purchases a lower supporting objective and an upper original-value enclosure. The missing step is to determine exactly which original action derivatives that information permits, before paying for a second action or a global feature cover. This section gives that conversion and a local policy safeguard. The calculation is available without alteration to every conventional representation in the comparison menu. Its contribution is an action-level use of already paid Bellman information, not a neural-exclusive property of smooth descent. Approximate supporting information and its accuracy–work tradeoff have a classical inexact-oracle interpretation (Devolder et al., 2014). Sharp smooth-convex information bounds also have an established interpolation literature (Taylor et al., 2017). The calculation here states exactly what the saved original-risk support licenses at a capacity-constrained action; it does not claim a new general theory of projected gradient methods.

Fix an original state x, put $A = A(x)$, and write $q = Q_x$ on $[0, A]$. The retained original-model result gives $\mu = 2$ and $L = L_\alpha$ in Theorem S.4; no fitted-critic curvature is used. Suppose a completed original certificate at $a \in [0, A]$ provides numbers

W, U, p_-, p_+ and an actual supporting slope $p \in [p_-, p_+]$ for which

$$q(a) \le U, \qquad q(b) \ge W + p(b-a) + \frac{\mu}{2}(b-a)^2 \quad (0 \le b \le A). \tag{20}$$

The slope need not equal $q'(a)$. In particular, the slope of an approximate terminal Bellman minorant is not differentiated as if its optimization error were zero. Put $e = U - W \ge 0$ and $c = (L-\mu)/2$. All these data refer to the original optimal-future objective. A saved upper bound on implemented-policy cost also bounds $q(a)$ and may be used conservatively. Subtracting the future-policy allowance is permitted only when the source formula establishes that the allowance was explicitly added to an otherwise valid q upper bound.

For $r > 0$ define the computable uncertainty function

$$\Psi(e,c,r) = \inf_{0<h\le r} \{e/h + ch\} = \begin{cases} 2\sqrt{ec}, & c > 0,\ e \le cr^2, \\ e/r + cr, & e > cr^2 \text{ or } c = 0. \end{cases} \tag{21}$$

The first expression has value zero at $e = 0$. When $r = 0$, set $\Psi(e,c,0) = +\infty$; absence of a feasible direction does not provide an absent derivative bound. A rational or outward implementation can use any feasible h in place of the infimum.

THEOREM 6—Exact derivative information in a paid quadratic support: *Suppose q is continuously differentiable, μ-strongly convex, and has L-Lipschitz derivative, and (20) holds. Then*

$$p_- - \Psi(e,c,A-a) \ \le q'(a) \le p_+ + \Psi(e,c,a). \tag{22}$$

Intersecting with any separately proved original-gradient enclosure remains valid. For the investment problem $[-7,7]$ is such an enclosure. For an exact supplied slope p and without that additional gradient information, both endpoints of (22) are sharp in the stated information class: each finite endpoint is attained by a function with constant curvature L, the same support, and $q(a) = U$. At a missing side, the corresponding derivative is unbounded in this class. Thus a method given only these saved data cannot replace a straddling derivative interval by a certified sign using its representation label.

The capacity asymmetry matters. At $a = 0$ the support supplies a lower bound on the inward derivative, which can certify optimality when positive; it does not supply

its upper bound. At $a = A$ the opposite statement holds. An arbitrary approximate supporting slope can have a different sign from the actual derivative at a capacity boundary. The supplement gives an exact rational counterexample and verifies the sharpness statements.

6.1. *A replacement which pays no second-action oracle when resolved*

Let $[g_-, g_+]$ be a valid finite version of (22). For a feasible displacement $d \in [-a, A-a]$, define

$$\overline{m}(d) = \max\{g_- d, g_+ d\} + \frac{L}{2} d^2, \qquad \underline{m}(d) = \min\{g_- d, g_+ d\} + \frac{\mu}{2} d^2. \tag{23}$$

The robust smooth-descent proposal is the minimizer of $\overline{m}$: it is zero if $g_- \le 0 \le g_+$, $-\min\{a, g_-/L\}$ if $g_- > 0$, and $\min\{A-a, -g_+/L\}$ if $g_+ < 0$. The executed finite menu uses the six multiples $1, 1/2, 1/4, 1/8, 1/16, 1/32$ of this proposal. Among the candidates passing all three tests below, select the one with largest checked lower gain, with ties resolved by the smaller realized action and then the menu index. Retain all six attempted calculations and their costs. A smaller step can pass the objective-width test when the unconstrained robust step cannot. This rule is optimal only over the recorded menu under that certified-gain criterion; it is not a global policy optimizer. The realized action must be feasible after rounding. Recompute the bounds at that realized action; an ideal unrounded minimizer does not license the rounded policy.

The actual future controller is unchanged and satisfies $J_x(b) = q(b) + \eta_x(b)$ with $0 \le \eta_x(b) \le \delta$ from (10). No regularity of η_x is imposed. For the realized $b = a + d$, put

$$\begin{aligned} U_b &= U + \overline{m}(d) + \delta, \\ L_b &= W + \min\{p_- d, p_+ d\} + \frac{\mu}{2} d^2, \\ B &= \min_{s \in \{p_-, p_+\}} \; \min_{u \in [-a, A-a]} \{W + su + \tfrac{\mu}{2} u^2\}. \end{aligned} \tag{24}$$

A stronger previously verified original-optimum lower bound can replace B by its maximum with that bound.

THEOREM 7—Original-policy gain and qualification from the same payload: *The constructed quantities satisfy*

$$J_x(b) - J_x(a) \in [\underline{m}(d) - \delta, \overline{m}(d) + \delta], \tag{25}$$

$$L_b \le J_x(b) \le U_b, \qquad 0 \le J_x(b) - V_0(x) \le U_b - B. \tag{26}$$

Consequently a strictly negative upper endpoint in (25) *certifies actual economic improvement. When in addition* $U_b - B \le \tau$ *and* $U_b - L_b \le \omega$*, the replacement satisfies the same complete-policy regret and objective-width requirements as the incumbent. These three tests use no new original-risk value at* b*. They do require the original paid payload, its checked smoothness bound, all interval arithmetic, action rounding, and their computational charges. If a test is unresolved, retain the qualified incumbent or enter the separately budgeted original-model fallback.*

A finite sequence of accepted replacements preserves feasibility and qualification, and the sum of certified gains $-\overline{m}(d) - \delta$ *lower-bounds the final actual gain. The statement is uniform in state where the premises hold; it is neither a fitted-loss statement nor a population adoption conclusion.*

For an interior anchor, an exact supplied support slope, and the exact untruncated interval in (22), the robust upper model is also sharp at each fixed displacement: the worst derivative endpoint and its constant-curvature-L witness attain it. This sharpness does not extend automatically to a wider outward or finite-probe enclosure, or after additional investment-specific information is imposed. Failure of the implemented finite menu therefore records an unresolved calculation; it does not prove that every method using the original formulas must refuse. A neural proposal can locate the first certificate evaluation, but a spline, polynomial, matched hinge, or online method receives exactly the same conversion and safeguard.

6.2. *Decision precision, finite refinement and priced stopping*

The precision of a supporting value and that of its action information have different units. At an interior action with $r = \min\{a, A - a\} > 0$ and $e \le cr^2$, the derivative interval width is at most

$$(p_+ - p_-) + 2\sqrt{2(L - \mu)e}. \tag{27}$$

This is the bound for the exact infimum in (21). A finite list of feasible rational h values remains valid but can be wider; its actual checked radii, including any smallest-step floor, must be used. Thus reducing a support-value error by a factor four reduces this exact component of action uncertainty by only a factor two. Conversely, a certified marginal tolerance ϵ requires both the slope enclosure and the support gap to be allocated explicitly; a small critic fitting loss does not supply either quantity.

Suppose a declared refinement ladder at this same action guarantees $e_k \le e_0 4^{-k}$ and slope width at most $w_0 2^{-k}$, at counted incremental work C_k. These are oracle assumptions to be verified for the implemented original-risk calculations. Put $D = w_0 + 2\sqrt{2(L - \mu)e_0}$. Once the interior condition is reached, the exact-infimum derivative width is at most $D2^{-k}$. A finite-radius implementation does not inherit this rate without a corresponding radius-approximation guarantee. If $g = q'(a)$ and

$$G_L(a, g) = \max_{d \in [-a, A-a]} \{-gd - Ld^2/2\},$$

the computed unrounded descent has guaranteed q improvement at least $G_L(a, g) - AD2^{-k}$. Hence a true smooth-model margin $G_L(a, g) > \delta + \gamma$ resolves its gain test once $AD2^{-k} < \gamma$, with total actual acquisition work $\sum_{j \le k} C_j$. Qualification width is an additional test. Even at exact derivative information, the Taylor interval can retain width $(L - \mu)d^2/2 + \delta$ at the proposed action. A ladder that cannot meet ω from that information must acquire a candidate value or fall back; it must not declare success from the gain test alone.

There is a narrow but exact priced stopping region. Suppose the current qualified policy has certified original regret at most K, one unit of additional computation costs λ units of original objective, and any allowed continuation of the solver must

first pay at least $c_* > 0$ new units. Exclude reuse of that information for other decisions from this single-decision comparison. If $\lambda c_* \geq K$, stopping is weakly optimal against every such continuation: even the exact economic optimum cannot save more than K. This is a dominance rule for a stated compulsory cost, not a general optimal-stopping theorem inferred from historical clock timings. Actual cost intervals, possible reuse, and the complete comparative menu remain part of an adoption study.

7. DIRECT STOPPING FOR THE ORIGINAL BELLMAN DECISION

A continuation critic is useful only if its proposal can be turned into a qualified economic decision at a competitive complete cost. The inherited warm service begins with an already completed conventional service and then purchases more original information. The direct service instead pays for its own actor proposal and its own original continuous-risk supports, and asks after each support whether the original decision is already resolved. It uses the same original model, continuation policy, candidate feasibility and qualification tolerances. Every representation receives this service, including the ordinary hinge with exactly the neural features and coefficients. The comparison therefore tests a decision procedure available to all representations; it does not assign its gains to neural architecture.

The stopping rule separates the action question from the absolute-value question. Let q be the original optimal-future action value, $q^* = \min_{[0,A]} q$, and $J = q + \eta$ the unchanged implemented policy, where $0 \leq \eta \leq \delta$. Original derivative intervals and the proved curvature bounds give a certified optimizer bracket $[\ell, r]$. At a realized feasible action a, write $\rho(a) = \max\{|a - \ell|, |a - r|\}$ and define the constrained projected gap

$$D_\mu(a; g_-, g_+) = \max_{z \in \{g_-, g_+\}} \left\{ -z d_z - \frac{\mu}{2} d_z^2 \right\}, \qquad d_z = \mathrm{proj}_{[-a, A-a]}(-z/\mu),$$

where $[g_-, g_+]$ encloses the original $q'(a)$. Retained supports also give $L_q(a) \leq q(a) \leq U_q(a)$ and $B \leq q^*$. Proposition S.44 proves that

$$J(a) - q^* \leq G(a) := \min\{U_q(a) + \delta - B,\ \delta + D_\mu(a; g_-, g_+), \delta + M\rho(a)\}.$$

When original endpoint derivative signs certify a stationary optimizer, $\delta + L\rho(a)^2/2$ is an additional route in this minimum. The projected bound also handles capacity optima with nonzero gradients. No derivative of the arbitrary future-controller error η is used.

Qualification requires both $G(a) \le \tau$ and the separate absolute interval condition $U_q(a) + \delta - L_q(a) \le \omega$, after outward rounding. The interval for the original optimum remains a distinct object: $J^+(a) - G(a)$ is not generally an optimum lower bound. A known optimal action can still have an unresolved additive value level, as Proposition S.45 shows. This separation identifies action sufficiency independently of the value refinement still required by the stated output; it does not assert that the finite implemented acquisition schedule minimizes either cost. It does not turn optimizer localization into free value information.

The first completed qualified action needs no comparison to a completed incumbent. Every later distinct-action replacement must itself meet both gates and have a strictly negative upper bound on the original complete-policy contrast. In particular, $\overline{\Delta}_q(a,b) + \delta < 0$ certifies a gain of at least $-\overline{\Delta}_q(a,b) - \delta$. A sharper certificate at the same action is recorded as requalification, not a policy improvement.

Further action centers use the actually rounded midpoint of the certified bracket. After the first actor center, moving to another center requires the observed contraction $w_{\text{new}} \le w_{\text{old}}/2 + \epsilon$, where ϵ is the actual rounding error relative to the old midpoint. The finite precision ladder and resource cap may prevent that condition or the stronger regret target from being reached. The service then retains its ordinarily qualified policy, or pays the complete conventional fallback if none qualifies. Lemma S.47 and Proposition S.48 prove these claims with the rounding and payment rules made explicit. Comparative cost or economic superiority requires the separately executed matched experiment.

7.1. *Exact reuse of certificate geometry within a paid decision*

An original-risk support can be expensive to acquire, but repeated manipulation of its certificate can also dominate a complete service. The transported lower and upper derivatives are piecewise-affine functions. Reconstructing their partitions

separately for every support anchor, candidate and contrast repeats a calculation determined entirely by the same paid information. We therefore examine a separately frozen implementation which retains this geometry within one paid decision. The first implementation and its measured costs remain distinct evidence.

For a fixed immutable record, let e_- and e_+ be the two derivative envelopes, and store their exact full-domain partitions together with

$$F_\pm(t) = \int_0^t e_\pm(u)\,du.$$

For $a < b$, the original contrast interval is $[F_-(b) - F_-(a), F_+(b) - F_+(a)]$; for $a > b$ the two endpoints reverse. A previously unseen endpoint requires a binary search and one rational quadratic evaluation. Previously evaluated endpoints, gradients and constrained envelope roots are retained. Requested subpartitions are obtained by clipping the full partition, with every original crossing retained. The raw-support minimum has a separate key containing its complete support levels and slope intervals. Candidate policy certificates are recalculated using the current incumbent, original-optimum bound and future allowance; none of those quantities is silently omitted from a cached policy result.

Proposition S.49 proves equality to every substituted original geometry calculation, including its returned subpartition list and boundary cases. Under the unchanged deterministic original evaluations and stopping rules, this yields the same economic queries, rejected candidates, policy decisions and certificates. The implementation also requires exact saved-record equality in its separately declared experiment; mathematical equivalence is not substituted for that execution check.

Proposition S.51 gives the precise work scope. With s retained derivative observations, each inherited partition has at most $1 + s + \binom{s}{2}$ pieces. Its original all-crossing construction takes $O(s^3)$ rational arithmetic and comparison operations. Once constructed, a new primitive or gradient argument takes $O(\log P)$ breakpoint comparisons for P stored pieces; a requested subpartition additionally pays for each piece returned. Each distinct immutable key incurs its own construction. Exact key formation, support-minimum requests, storage and cache management remain

charged. These are operation bounds for the stated certificate calculation, not a bit-complexity bound, an uncached lower bound, or a theorem about elapsed time.

Every neural and conventional service receives the same implementation. The cache is destroyed after each individual decision, including exception paths. It reuses no economic observation, terminal solve, actor output or completed service from another paid decision. Thus any measured improvement belongs to the common decision service under the same information, and its economic value still depends on complete costs and the original decision accuracy. The new paired study retains slower outcomes as well as faster ones and does not reinterpret unchanged policies as new policy gains.

8. COMPUTING THE DECISION CONTENT OF PAID BELLMAN INFORMATION

The complete-support characterization identifies the information in a Bellman calculation, but an unverified convex-program output does not yet give an economic certificate. We now give a finite calculation which can stop at any resource limit with a valid original-action bound. It combines tangents taken at different points, verifies their aggregate by rational arithmetic, and retains the stronger of this calculation and the already available conventional bounds. Cutting-plane optimization is classical (Kelley, 1960); smooth-convex interpolation and information-based worst-case calculations also have an established literature (Taylor et al., 2017, Florea and Nesterov, 2024). The contribution here is the explicit passage from the original, globally valid Bellman support record to a finitely verified action and regret calculation. Its economic information and certification rights are identical for a neural critic, a matched ordinary hinge, a polynomial, a spline, or an online solver supplied with that record.

8.1. *A finite certificate from numerical cut proposals*

Fix the state, risk law, feasible interval, original optimal-future objective, and paid record of Section 25. Write its nonempty finite-dimensional compatibility set as

$$P = \{z \in \mathcal{B} : c_i(z) \le 0,\ i = 1, \dots, m\}, \qquad \mathcal{B} = \prod_k [\underline{z}_k, \overline{z}_k].$$

The box is finite and rational. Linear restrictions, including equalities written in both directions, are among the c_i. For a requested original contrast $q(b) - q(a)$, let $H(z) = h_0 + h^\top z = q(a) - q(b)$. At arbitrary rational points v_r, choose valid rational subgradients $s_r \in \partial c_{i_r}(v_r)$ and store the exact affine cuts

$$A_r = s_r^\top, \qquad b_r = s_r^\top v_r - c_{i_r}(v_r), \qquad A_r z \le b_r.$$

The points need not be feasible, need not coincide, and need not be the solutions of an optimization problem. All nonlinear support constraints have explicit rational subgradients; the supplement gives the formulas.

THEOREM 8—A finite verified bound from a shared cut record: *For any nonnegative rational weights y_r, define*

$$R = h + A^\top y, \qquad B = h_0 - y^\top b + \sum_k \min\{R_k \underline{z}_k, R_k \overline{z}_k\}. \tag{28}$$

Then every compatible original objective satisfies $q(b) - q(a) \le -B$. Neither primal feasibility, stationarity, strong duality, nor a successful solver exit is required. A numerical linear program may propose the weights; sign checks, rationalization, reconstruction of each exact cut, and evaluation of (28) determine whether a certificate is accepted.

If a rational point $z^{\mathrm{f}} \in P$ is independently verified, then

$$-H(z^{\mathrm{f}}) \le U_{\mathrm{full}}(a,b) \le -B, \qquad 0 \le -B - U_{\mathrm{full}}(a,b) \le H(z^{\mathrm{f}}) - B. \tag{29}$$

Thus a checked gap can certify accuracy relative to the exact declared information bound. Without such a gap, the returned upper bound is valid but is not certified to be the sharp information bound.

Intersecting each accepted bound with all earlier valid bounds for the same original contrast gives a nonincreasing sequence of upper bounds. The cut record may be reused for several linear contrasts on one common node set, provided all arithmetic and optimization work is charged.

The zero-weight choice supplies a valid, possibly weak box bound. A failed optimization, an unhelpful rationalized dual, or the imposed cut limit therefore cannot invalidate the incumbent certificate. Feasibility of the original information

class is a scientific premise established by the original calculation; it is not inferred from a numerical solver's message. If an exact constraint check detects inconsistency in a supplied record, the record is rejected rather than used to prove a favorable decision.

The finite computation differs from an asymptotic convergence claim. With exact linear subproblem solutions, bounded subgradients, and a cut for every convex constraint at every current minimizer, the nested outer relaxations converge in value to the exact compatibility program. A proof is supplied in the supplement. A capped floating-point implementation uses only (28); it does not inherit that limiting statement as a finite accuracy certificate.

The information boundary also governs comparisons. The cheap conventional calculation clips transported derivative envelopes by the same original cap $[-M, M]$ and uses every available original-optimum lower bound in its endpoint enclosure. If such an optimum bound has not been encoded as a constraint of P, intersecting with it remains valid for the actual original objective, but is not a tighter numerical solution of U_{full}. The raw cut bound and the final original-policy intersection are reported separately. No information omitted from a comparator is attributed to the convex solver or to neural architecture.

8.2. *From pairwise contrasts to original-policy regret*

The same record can answer a stronger economic question: how much original value could still be lost by returning a particular action? Define its information-compatible regret by

$$\Gamma(a) = \sup_{q \in \mathcal{Q}_{\text{full}}} \{q(a) - \min_{x \in [0,A]} q(x)\}. \tag{30}$$

It is a property of the declared original information. It is not a fitting loss, a posterior mean, or a claim about a different recursive preference. Introduce an action grid $0 = x_0 < \cdots < x_n = A$ containing a, with maximum spacing h. These are additional unknown interpolation nodes, not newly purchased economic observations. Theorem S.39 ensures that the augmented finite program represents the same function class.

THEOREM 9—Finite decision resolution under unchanged information: *Put* $u_j = U_{\text{full}}(x_j, a)$. *Then*

$$\max_j u_j \le \Gamma(a) \le \max_j u_j + \frac{Lh^2}{8}. \tag{31}$$

The constant $1/8$ *is sharp, even for a compatible class consisting of one rational quadratic. Both endpoints must be in the grid; a constrained boundary optimum is then represented exactly.*

More generally, let $\widehat{u}_j \ge u_j$ *be any verified upper contrasts, setting* $\widehat{u}_j = 0$ *at* $x_j = a$. *For a cell of length* $d_j = x_{j+1} - x_j$, *put*

$$G_j(a) = \max_{0\le t\le 1} \left\{ (1-t)\widehat{u}_j + t\widehat{u}_{j+1} + \frac{Ld_j^2}{2} t(1-t) \right\},$$

$$G(a) = \max_j G_j(a). \tag{32}$$

Then $0 \le \Gamma(a) \le G(a)$. *For rational inputs* $G(a)$ *is an exact rational number: with* $k_j = Ld_j^2/2$, *the maximizing argument is*

$$t_j^* = \text{proj}_{[0,1]} \frac{\widehat{u}_{j+1} - \widehat{u}_j + k_j}{2k_j}. \tag{33}$$

If every $\widehat{u}_j - u_j \le \epsilon$, *then*

$$0 \le G(a) - \Gamma(a) \le \epsilon + \frac{Lh^2}{8}. \tag{34}$$

Every quantity here concerns the same original objective and the same paid record. Refining the computational grid requires additional certificate work, but no new risk or continuation-value evaluation.

For the unchanged implemented future $J(a) = q(a) + \eta(a)$, $0 \le \eta(a) \le \delta$, the returned action consequently has original-policy regret at most $G(a) + \delta$. Intersect this with every earlier valid regret bound. An absolute original-value interval at a is still required separately: grid resolution cannot identify a free additive value level. If a policy is changed, a strictly negative upper bound on the actual contrast

$J(b) - J(a)$ is still required. A reduction in worst compatible regret by itself does not prove a strict gain for the realized economy.

There is also a finite robust-action implication. The function Γ is M-Lipschitz and μ-strongly convex. If the same grid is used for candidate actions, and the verified pairwise accuracy is at most ϵ for every candidate, a minimizer $\widehat{a}$ of the computed G satisfies

$$\Gamma(\widehat{a}) - \min_{a \in [0,A]} \Gamma(a) \leq \frac{Mh}{2} + \frac{Lh^2}{8} + \epsilon. \tag{35}$$

This is an information-compatible minimax regret guarantee, not identification of the actual optimal action and not an advantage due to a neural encoding. The supplement states its proof and scope.

Finally, suppose any further computation for this one decision must first cost at least $c_* > 0$ resource units, priced at λ units of original objective, and exclude reuse for other decisions. Stopping is weakly optimal against all such continuations whenever

$$\lambda c_* \geq \min\{K_{\text{old}}, G(a) + \delta\},$$

where K_{old} is a previously valid original-policy regret bound. This is a dominance conclusion conditional on an actual compulsory cost and the declared original objective. It does not turn observed host clocks into a deployment probability or an estimated economic price.

9. A PROSPECTIVE ORIGINAL-ECONOMY COMPARISON

9.1. *Economic object and common information*

The experiment uses exactly the two-dimensional primitives in Section 5, with $\alpha = 1/2, 3/4$ and the same implemented continuation controller. Write q_x for the optimal-future Bellman action objective and J_x for the complete implemented policy value. The proved relation is $q_x \leq J_x \leq q_x + \delta$, with $\delta = 15/2^{24}$. An ordinary service must have original-policy regret at most $1/256$ and an absolute original-value interval no wider than $1/8192$. The separately reported stronger regret target is $1/65536$. Neither passing the ordinary gate nor narrowing a residual constitutes success at that stronger target.

All nine states in $\{3/16, 9/16, 15/16\}^2$ enter in lexicographic order. Every coordinate is absent from both acquired training meshes. These are fixed test locations, not random draws from an economic state distribution. The complete executable closure of 538 scientific files and the three protocols were remotely frozen at 769e3eb3ac08beaa7bb6778c1676f15e1f8d6d2b. All 543 published file objects were read back before the first economic calculation. The source manifest and execution receipts identify this study separately from the preceding four-state experiment and protected independent studies. The states and service parameters were chosen after examining the published prior evidence, and before observing any present outcomes.

The 56 actors retain acquisition commit 796c5f2badf68b051ac6c144884c0e61b87faf5e. For each risk, both five- and nine-node coordinate meshes are used, with quadratic, ReLU, matched native hinge, spline, degree-two polynomial, degree-three polynomial and grid representations, each fitted by value or by original marginal information. Their 25 and 81 training states receive 13 original-objective contexts each. Every representation at a given risk and mesh has access to the same acquired labels. The hinge copy pays its required ReLU acquisition parent; its matched features and coefficients are checked exactly. Historical acquisition allocations remain historical measurements and are not replaced with new clocks.

Each actor and each risk-specific original online controller is served by three independently executed complete routes. The fused control is the retained strong service including qualification, rejected finite-menu proposals and fallback. The warm adaptive route first pays that whole service and then buys additional action information. The direct terminal route pays only its own actor proposal and initial original support before attempting qualification; it receives no completed prefix for free. A 33-action original-law dynamic-programming service is included at each risk. The catalogue therefore has 176 cohorts and 1,584 primary decisions.

9.2. *Paid refinement and unchanged-policy improvements*

The direct route starts at 32 risk bins. Subsequent per-center precisions are $32, 64, 128, 256, 512, 1024, 2048$. A query at n bins requests $2n + 1$ terminal mid-

point or endpoint contexts, with all actual terminal iterations and risk calculations recorded separately. The limits are eight centers and 32,768 requested contexts, checked before acquisition. Rejected levels and unsuccessful candidates remain charged. Subsequent centers use the midpoint of a certified optimizer bracket, rounded down on the inherited 30-bit feasible action lattice; contraction tests include the actual rounding displacement. Fixed caps need not attain an asymptotic theoretical allocation.

At every saved candidate, the service intersects the valid level-gap, projected-gradient and terminal-bracket regret accounts. The first ordinarily qualified candidate is chosen by direct regret, then upper value and action. After that point, changing the action requires an independently valid upper original contrast plus δ to be strictly negative, as well as both ordinary qualification gates. An unchanged action may receive a sharper certificate without being counted as a gain. The service stops when its returned action meets the stronger regret target and the separate value-width gate. If the cap expires, it keeps a qualified incumbent; if none exists, it pays the full fused fallback, retaining all failed direct work. Infrastructure or source-identity errors abort and preserve their partial outputs.

The support-certificate subexperiment is separately fixed to 74 paid pools: all 58 first-state fused services and the 16 remaining state–risk online services. Its candidate actions are the realized incumbent and the capacity boundary. Five equally spaced action nodes, augmented by the incumbent, control continuous-action regret. Each pairwise problem permits at most eight linear-programming calls, with at most 2,000 iterations per call; the initial all-constraint tangents are followed by at most 32 exact most-violated cuts per round. Points and dual proposals are rationalized on a 2^{-36} lattice. Every proposed cut and weight is retained. The cheap comparator uses the same paid supports, clipped derivative cap and original-optimum lower bound. This experiment purchases no additional economic evaluations. Its original prefix cost and added numerical and rational-algebra work are both charged. It does not rewrite any primary service policy after seeing the results.

9.3. *Independent evidence and fitting audit*

The saved-arithmetic audit replays feasibility, budgets, every incumbent change, exact regret algebra, the matched ReLU/hinge identities, and all timed scientific repetitions. A separate rational implementation reconstructs the original continuous-risk economics without importing the production compiler, actor, quadrature or certificate code. Every distinct final state–risk–action key is evaluated at eight bins, followed in fixed hash order by 32 and 128 bins while unresolved and within 32,768 requested contexts. Terminal root bisection uses 32 iterations. An audit that cannot cover all keys at its initial level fails; precision is not extended after observing outcomes. Complete-policy identities imply zero contrasts exactly. Otherwise independently bounded improvements, losses and unresolved comparisons are all reported.

Value-interval containment, confirmation of the claimed regret, ordinary and stronger qualification, and contradictions are separate outcomes. A looser independent regret bound is unresolved. A disjoint value enclosure or a proved regret lower bound above the production upper claim is a contradiction. The independent evaluator encloses the optimal-future objective and uses the proved same-controller allowance; it does not replay every terminal implementation branch throughout the continuum. The full witnesses and bisection ledgers are retained. Independent standard-library rational code also reconstructs each support-program cut, residual, weight and final bound without importing its production solver or scientific libraries.

The separate acquisition audit uses the fixed 181-file historical input closure. It independently rebuilds feature matrices, including ordinary hinges, spline recursions and original-action derivatives, then solves 48 finite-design fits. The eight matched hinge models are accounted for as copies, not additional independent fits. Value fitting uses an independently assembled bounded-variable least-squares problem configured for BVLS instead of TRF. Four actual solves execute the different active-set routine; 20 return the feasible unconstrained solution through the shared initial numerical path. Four VALUE hinge readouts copy their paid ReLU solution; marginal fitting uses independently assembled shifted robust linear programs with

the shared HiGHS library. Exact rational lower bounds and coefficient rounding allowances audit the finite-design objectives (Supplement, Section 32). No refitted actor replaces a frozen production actor. The audit does not acquire fresh labels or identify a population neural advantage.

9.4. *Complete costs and economic adoption*

Three serial processes each repeat the entire 176-cohort catalogue at $(3/16, 3/16)$, for 528 further decisions. Their order rotates by $13r$ in repetition $r = 0, 1, 2$. Models, actions, certificates, operation counts and stop reasons must reproduce the corresponding primary records. Each clock includes its service and output, together with an equal allocation of the measured parent residual. Thread counts are one; host load and clock frequency are uncontrolled. The observed range is descriptive, not a sampling confidence interval or a portable operation-time conversion. Source verification, independent economics, fitting, solver certificates and reporting have separate research-work receipts. Unknown interrupted historical parent costs remain unknown.

For each ReLU service, risk and $N \in \{1, 32, 1000, 10^6\}$, adoption is assessed against every conventional service through $J + \lambda\{t + K/N\}$, where t is complete first-state decision time, K is its retained acquisition allocation and $\lambda \geq 0$ prices a second in objective units. Sufficient regions use own upper versus rival lower bounds; optimistic outer regions reverse them. Exact identical-policy blockers are also reported without replacing the interval comparisons. Economic values from the other eight states are not paired with first-state timing. These prices and amortization counts are explicit sensitivity scenarios, not an estimated value of computation or a calibrated adoption forecast.

10. ORIGINAL DECISIONS AND COMPLETE COMPUTATIONAL WORK

All 1,584 predeclared primary decisions and all 528 timing repetitions executed successfully. The saved-arithmetic audit passed 30,703 checks, including 4,537 candidate-certificate replays, 1,805 direct-query attempts, 216 exact ReLU/hinge policy-and-work identities, and every timed scientific outcome. A separate exact

TABLE I

COMPLETE ORIGINAL-ECONOMY SERVICE CENSUS

Service	Decisions	Ordinary	Stronger	Terminal rows	Service seconds
Fused control	522	522	296	902,961	14.573
Warm adaptive	522	522	522	973,410	30.825
Direct terminal	522	522	522	418,957	63.009
Action-grid DP	18	18	15	176,223	7.819

Ordinary regret is at most 1/256, stronger regret at most 1/65536, and absolute value width at most 1/8192. Terminal rows report one original-model work counter; other operation counts remain in the ledger, and certificate algebra is included in measured service costs. The displayed seconds sum the complete primary service clocks. Whole parent, input, output and inter-cohort overhead have separate receipts and are allocated in the timing and adoption comparisons.

squaring audit verified all 2,112 reported action-distance upper bounds. These are upper bounds on action error, not measurements of the unknown exact error.

The direct service and warm adaptive service meet the stronger target in every one of their 522 decisions; the fused control meets it in 296. All 18 action-grid decisions qualify ordinarily, while 15 meet the stronger target. No direct fallback or recoverable numerical failure occurs. The direct service records 689 strictly certified internal replacements after an incumbent has qualified. These are sequential improvements over that incumbent, not 689 wins over the fused or warm comparator.

The primary measured service-cost result is unfavorable to the uncached direct implementation. It uses 418,957 compiled terminal rows compared with 973,410 for warm adaptation, a 57.0 percent reduction, but its measured service component consumes 63.009 seconds compared with 30.825 seconds. Against the fused control, it achieves the stronger target more often while using fewer terminal rows, yet also takes more service time. The source report retains model loading, row output and whole-process residuals separately; complete first-state and paired-study accounts include these costs. Allocating the first primary parent residual equally across all 176 cohorts gives whole-process totals of 15.559, 32.377, 64.328 and 7.854 seconds for fused, warm, direct and grid services, respectively; the saved cost reconciliation does not retime any decision. Exact certificate work is a material input: reducing integration work alone does not establish computational competitiveness. The

TABLE II

ORIGINAL POLICY VALUES AND REGRET CERTIFICATES ON THE NINE-STATE CENSUS

Service	Mean original cost interval	Mean regret bound	Maximum action bound
Fused control	[0.56303588, 0.56304558]	0.00060365	0.0620625
Warm adaptive	[0.56267474, 0.56267703]	0.00000517	0.0038902
Direct terminal	[0.56267442, 0.56267856]	0.00000623	0.0038986
Action-grid DP	[0.56267659, 0.56267767]	0.00000729	0.0053816

Policy intervals and nonnegative bounds are rounded outward. The first three rows each cover the identical 58 actor–risk cohorts at all nine states; the grid row contains the two original-risk controls at those states. No population weighting or monetary calibration is asserted.

original primary observations are retained when the distinct cached-geometry algorithm is assessed below.

10.1. *Independent original economics and adoption conditions*

The independent continuous-law evaluator covers all 515 distinct final state–risk–action keys and uses 32,740 of its 32,768 allowed terminal contexts. Its 884 evaluations and 915,872 actual bisection iterations are additional research work, with an internal elapsed time of 491.7335 seconds. It independently establishes the ordinary regret threshold in all 1,584 decisions and the stronger regret threshold in 1,175. It confirms 1,277 of the specific reported regret bounds and verifies 471 reported value intervals by containing the independently reconstructed interval inside the reported one. The remaining 307 regret checks and 1,113 interval-containment checks remain unresolved under the fixed cap. A separately recorded post-execution exact audit of the already saved independent bounds verifies the width gate in all 1,584 cases; their maximum independent width is below 0.000107509. Thus the combined independent bounds qualify all ordinary services and 1,175 stronger services. This additional saved-bound check leaves the frozen regret-only summary fields unchanged and performs no further economic evaluation. There are zero contradictions. These counts describe different tests; an ordinary regret-threshold pass does not confirm every tighter production claim or by itself recheck its separate value-width gate, and an overlapping interval is not containment.

There are 1,044 paired comparisons of the direct route against the fused and warm route for the same actor, risk and state. Independent actual-policy bounds certify 238 direct gains and 39 direct losses, establish 517 exact complete-policy ties, and leave 250 orderings unresolved. A stronger regret upper bound therefore does not guarantee that a service beats a particular competing feasible policy. Against the fused control separately, there are 214 gains, 13 losses, 260 exact ties and 35 unresolved comparisons; the outward fixed-census mean contrast is contained in $[-0.0003681, -0.0003530]$. Against warm adaptation, there are 24 gains, 26 losses, 257 ties and 215 unresolved cases, with mean contrast contained in $[-0.000007002, 0.000007125]$. Thus an aggregate original cost improvement over fused control is certified, while the mean comparison with warm adaptation remains unresolved. The full split and all unresolved pairs remain in the companion and machine-readable ledger. These deterministic results carry no population confidence interval.

The three complete first-state timing processes total 17.7213 seconds. In 68 of 116 matched first-state comparisons, the direct route's largest measured complete time is below the comparator's smallest; none satisfies the reverse strict range separation. The other comparisons have overlapping ranges. This first-state observation coexists with the unfavorable nine-state total above; it must not be generalized to all states. Model loading, output and the whole-process residual are included in these timing ranges.

All 96 prespecified neural adoption scenarios have empty sufficient regions. All 96 optimistic interval-only outer regions are nonempty, but each scenario has an exact identical-policy conventional blocker whose worst observed total cost, including the retained acquisition allocation, is below the neural service's best observed total cost. Conditional on those recorded cost ranges, every strictly positive computation price excludes that neural service; at zero price it ties that particular blocker. This exact-policy diagnostic preserves the interval classifications rather than replacing them. No neural adoption, calibrated resource price or economic demand estimate is inferred.

TABLE III

COMPLETE-SUPPORT CALCULATION AGAINST THE SAME-INFORMATION CONVENTIONAL BOUND

Declared calculation or verified outcome	Count
Paid information pools	74
Pairwise original-action contrasts	888
Strictly smaller upper contrasts	48
Candidate-action regret bounds	148
Strictly smaller regret bounds	6
New ordinarily qualified strict policy gains	0
Additional original-economic queries	0
Linear-programming calls, including unhelpful proposals	5,920

10.2. *Full-support computation under identical paid information*

The finite solver processed all 74 predeclared paid pools, 148 candidate actions and 888 pairwise contrasts. It made 5,920 capped linear-programming calls and no additional economic query. There were no failed pools or reported LP failures. Its whole parent-process cost was 20.2483 seconds, additional to the original prefixes; the shared host was not controlled. An independent implementation reconstructed all 33,331 saved cuts and 1,508 nonzero rational weights, and verified every contrast and continuous-action regret aggregation. The 1,924 record checks took 4.2228 seconds internally, with their separate parent receipt retained.

The 48 improvements establish that the complete compatibility restrictions can sharpen a cheap, equally informed conventional bound in executed original model records. They do not establish a decision gain. All six smaller regret bounds concern the capacity candidate in online-control pools away from the first state. The improved bounds remain far above the ordinary service tolerance. No candidate becomes a newly qualified strict improvement and none changes the primary policy. The smaller contrast or regret is therefore an information improvement with positive computational cost in this study. The full-support calculation has not earned an economic adoption claim here.

The comparison already gives the conventional method the global derivative cap and the paid original-optimum lower bound. The raw cut certificate bounds

the complete-support information class. Its final intersection with the conventional bound is an original-policy bound and is not labeled a sharp solution of that class when the optimum floor was not encoded in its defining constraints. No independently feasible primal gap was supplied by this capped run; neither exact finite optimality nor achievement of the ideal limiting solution is claimed. All candidate failures, unchanged-policy outcomes, exact witnesses and algebra costs remain available in the development account.

10.3. *Independent acquisition implementation audit*

The separate historical-data audit reproduces all 56 stored acquisition cases: 24 value fits, 24 robust marginal fits and eight matched hinge copies. It independently reconstructs the fixed features, exact designs, one-sided original-action derivatives, dyadic rounding and serialization. The BVLS configuration executes a distinct active-set path for four value fits; 20 return the feasible unconstrained solution through the shared numerical initial-solve path. All 28 value-loss rounded readouts, including four hinge copies, reproduce the historical coefficients exactly. Overall 39 readouts match; the 17 differing marginal readouts are retained. All 56 cases pass the specified checks. The maximum exact finite-box sum-of-squares optimality-gap bound is below 2.34×10^{-4}; the maximum new robust marginal gap is below 2.84×10^{-8}.

These results address independent acquisition implementation without changing the deployed actors. The marginal formulation is independently assembled but shares the HiGHS solver; no external solver-implementation independence is claimed. The audit does not reacquire economic labels or prove approximation error on the entire continuous domain. It uses zero new economic queries and adds 21.824 seconds of complete research verification on the shared host. The proofs and full case records appear in the supplement and development companion. Finite fitting optimality and original-policy regret remain separate quantities.

The unchanged production report records independent_refitting_executed=False for its own service scope. The separately identified audit reported here has its own

TABLE IV

INDEPENDENT FINITE-DESIGN FITTING AUDIT

Representation	Same / 8	VALUE gap	Stored M gap	Audit M gap
quadratic	8	2.05×10^{-4}	4.30×10^{-9}	4.30×10^{-9}
relu	4	9.93×10^{-5}	3.26×10^{-9}	3.69×10^{-9}
hinge	4	9.93×10^{-5}	3.26×10^{-9}	3.69×10^{-9}
poly2	8	2.05×10^{-4}	4.30×10^{-9}	4.30×10^{-9}
poly3	7	2.34×10^{-4}	9.69×10^{-9}	6.50×10^{-9}
spline	4	1.35×10^{-5}	2.58×10^{-8}	2.84×10^{-8}
grid	4	6.27×10^{-10}	2.18×10^{-8}	2.30×10^{-8}

All four risk/budget cells per loss are retained. Positive decimal gaps are rounded upward. VALUE denotes sum squared loss; M denotes the robust marginal-node residual. VALUE coefficients coincide in every case. These are finite-design optimization gaps and do not measure economic policy regret.

execution, source-gate and research-cost receipts; it is not silently reassigned to that production process.

10.4. *Exact geometry reuse and complete computational cost*

After the initial excess cost of direct service was observed, a distinct algorithm was frozen prospectively at 9c6246ecad31acd6a937a2d8ad4710bcabc8c801. It reuses exact derivative-envelope geometry within each paid decision. Each full-domain partition, cumulative rational integral, gradient and constrained root can serve repeated candidate contrasts; the cache is destroyed at the end of that decision. Every conventional representation and strong control receives the same mechanism. The new experiment executes both the uncached and cached algorithms afresh, in fixed alternating cohort order, over all nine states and three further first-state timing processes.

All 3,168 new primary decisions and 1,056 new timed decisions completed. The exact verifier matched every one of their 4,224 complete scientific records to the first study after removing only declared clocks and run metadata. This checks the acquired supports, rejected proposals, selected actions, certificates, original-model

TABLE V

FRESH PAIRED COMPLETE COST OF EXACT GEOMETRY REUSE

Service	Decisions	Uncached s	Cached s	Reduction (%)
Fused control	522	16.247	15.930	1.9
Warm adaptive	522	35.106	27.417	21.9
Direct terminal	522	68.849	33.512	51.3
Action-grid DP	18	7.797	7.785	0.2

The table allocates the whole paired primary parent process, including service execution, model loading, row serialization and an equal share of the residual overhead. Uncached and cached columns are fresh observations under the second source boundary. The first study in Table I remains unchanged; its displayed service component and this whole-process allocation have different accounting scopes. Shared-host clocks are descriptive. All original policy outcomes and economic-operation counts are identical between the paired variants.

operation counts and stopping outcomes. Economic information is purchased again in each paid service; only exact algebra is reused within it.

Exact geometry reuse reduces the direct route's fully allocated nine-state cost by 51.3 percent, from 68.849 to 33.512 seconds. The same optimization also helps the warm control. Cached direct still costs 22.2 percent more than cached warm, whose complete cost is 27.417 seconds, while both meet the stronger target in all 522 decisions. The substantial reduction in certificate work therefore improves the direct implementation without establishing a universal advantage over the strongest service.

For the direct primary corpus, 49,376 contrast requests use 3,610 full-domain partition constructions and 7,930 new endpoint antiderivative evaluations; 131,514 additional primitive requests reuse exact values. These diagnostics account for algebraic reuse. The original 418,957 compiled terminal rows remain fully charged and unchanged.

The first-state timing workload yields a different comparison: cached and uncached ranges are strictly separated in favor of caching in 1 of 176 cohorts and against it in 1; the other 174 overlap. Cached direct has a strictly smaller observed range than its matched cached fused or warm control in 64 of 116 comparisons,

with 0 reverse separations. These fixed-state ranges are not population confidence statements and do not supersede the nine-state totals.

The cost calculation is repeated against conventional services that receive the same cache. All 192 sufficient neural adoption regions remain empty, all 192 optimistic interval-only regions remain nonempty, and all 192 scenarios retain an exact identical-policy conventional cost blocker. The computation improvement is shared across representations. The original independent audit's 307 unresolved regret checks, 1,113 unresolved containment checks and 250 unresolved policy orderings remain unchanged, because the cached experiment returns exactly the same complete policies and certificates.

11. ECONOMIC INTERPRETATION AND THE CONTINUING NBO CONTRIBUTION

The decision output of NBO is a feasible investment with its continuation controller under a specified recursive preference. The relevant loss is the value of that complete policy relative to the original optimum. A useful learned representation must reduce this loss, or reduce the complete resources required to reach a common economic tolerance. Those requirements motivate the three current methodological results: action-aligned stopping, verified use of the complete paid support record, and reuse of exact certificate geometry. They retain the original Bellman problem rather than optimizing a different fitting criterion and calling the result an economic improvement.

The combined evidence shows why all three accounts are necessary. Production certificates meet the declared stronger regret target throughout the 522-case direct-service census. The independent comparison establishes a lower mean original policy cost relative to the completed fused control by a certified amount between 0.0003530 and 0.0003681 on this fixed equally weighted census. There are nevertheless 13 individually certified losses against that control, and the stronger warm adaptive comparator has an unresolved mean comparison, with 24 direct gains and 26 direct losses. These are objective units of the stated investment model, not a measured currency benefit or a population treatment effect. Neither a lower upper bound on regret nor an internal accepted gain determines the ranking of two completed competing policies.

The uncached computation further shows that economic information can be purchased cheaply but processed expensively. Its substantial reduction in terminal integration rows comes with a larger measured service-time bill than warm adaptation. Exact support geometry was repeatedly rebuilt for questions that shared the same paid information. The separately prospective cached algorithm is motivated by that observed mechanism, retains the first result, and gives the same optimization to every conventional and neural service. Its complete paired costs and full scientific equality checks determine the scope of its numerical improvement. Classical numerical implementation remains a substantive comparator even when the actor is represented by a neural network.

The full-support solver answers a different question. It can exploit constraints that a cheaper derivative or endpoint calculation leaves unused, and its exact cut certificate retains validity even at an unsuccessful numerical exit or a finite iteration cap. That additional information tightens 48 executed contrast bounds and six candidate regret bounds. The tested candidate policies still fail to acquire any new qualified strict gain. The extra LP and rational arithmetic therefore have a measured bill and no additional realized policy benefit in that study. This is a useful boundary for algorithm design: solving a more complete information problem is economically warranted only when its remaining decision value can justify its full cost. The conditional stopping proposition makes that requirement precise without inventing a calibrated price of computation.

Independent fitting addresses another part of the method. It reconstructs finite-design objectives from the acquired labels, solves them through separately assembled numerical routes, and verifies rational objective and rounding bounds. Nonunique robust fits can have different coefficients without contradicting the same finite-design objective. None of these facts makes a finite training design into a uniform approximation theorem or supplies new population policy evidence. The original actors are retained, and the shared HiGHS library, unchanged labels and fixed features delimit the independence actually achieved.

Exact ReLU and native hinge equivalence is equally consequential. With the same features, readout, labels, feasible decisions and certification service, they implement the same economic rule. Their shared geometry cannot establish a neural-specific

advantage. The original 96 sufficient neural adoption regions are empty and have identical-policy conventional cost blockers under the recorded full cost ranges. The follow-up applies its implementation change to both sides of that comparison and reports its own adoption conditions. A competitive neural contribution must survive these comparators, as well as appropriate analytic, semianalytic, spline, polynomial and dynamic-programming methods with their actual information and acquisition routes charged.

The mathematical scope is broader than this one computation but remains conditional on explicit premises. The own-future construction and recursive error accounts apply across their stated finite-horizon economic domains. The stochastic-control and nonlinear-recursive-utility bridges require their own drivers, filtrations, existence and boundary conditions. The complete support program and its decision-resolution bounds concern scalar, smooth, strongly convex original action objectives under a specified information interface. They do not exclude algorithms using additional primitive formulas, prove a randomized multivariate information lower bound, or establish neural representation superiority at a common unrestricted resource budget.

The numerical application has two decision dates, two state coordinates, one constrained control and one continuous innovation factor. Its state mesh, risk levels and primitives are fixed; it is not newly calibrated to a firm or a market. A substantive next extension must demonstrate action-relevant information acquisition and complete decision value as these dimensions, horizon and constraint interactions grow. A new learned actor's population claim requires its own independent sample, and a practical adoption claim requires an explicit decision workload and defensible resource value. The present algorithms, proofs, reproducible positive and negative results, and protected historical protocols provide the basis for those further tests within the Neural Bellman Operators programme and its Econometrica objective.

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